Does the Cost of Being Wrong Constrain AI Adoption? Evidence from Anthropic API and Claude.ai Task-Level Data

Key Findings

- Occupations in the highest quintile of O*NET's Consequence of Error scale show task automation penetration rates of just 4.4 percent. That is less than one-third the rate of the lowest-consequence quintile at 16.1 percent.
- The gradient is monotonic across all five quintiles and holds in both Anthropic API traffic, which reflects firm-level automated workflows, and Claude.ai traffic, which reflects direct human usage. The correlation between consequence of error and adoption is -0.286 in API data and -0.302 in Claude.ai data.
- Rank correlation between the two platforms is 0.825, indicating that the occupations firms choose to automate and the occupations workers choose to use AI for are driven by the same underlying task characteristics.
- These patterns are consistent with a structural brake on AI adoption: where errors carry serious consequences, both firms and individual workers appear reluctant to delegate tasks to AI systems.

Introduction

The capabilities of large language models have advanced faster than their adoption in enterprise workflows. Models that demonstrably match or exceed human performance on coding, legal drafting, financial analysis, and clinical reasoning benchmarks remain, in practice, unevenly deployed across the economy. This incongruence is puzzling if one assumes that task performance is the primary driver of adoption. It suggests that performance is necessary but not sufficient, that other structural factors shape where and how quickly automation takes hold.

One such structural explanation is that automation is self-limiting in high-stakes environments, where errors carry serious professional, financial, or physical consequences, the cost-benefit calculus of delegating to an AI system shifts unfavorably.

The mechanism has a straightforward economic interpretation. A worker or firm deciding whether to delegate a task to an LLM weighs the expected gain from automation against the expected cost of errors. That cost is a product of two terms: the probability of an error and the consequence of that error. Current LLMs have broadly similar error rates across task types, a model that hallucinates a citation in a marketing report and hallucinates a drug dosage in a clinical note may be wrong at similar rates. But the consequence of the second error is orders of magnitude larger. As consequence of error rises, the expected cost term dominates the calculation even if error probability stays constant, and delegation becomes increasingly unattractive.

A second, subtler mechanism involves the variance of errors rather than their expected cost. In high-consequence environments, a risk-averse worker or firm penalizes not just the

expected cost of an error but the unpredictability of when and how errors occur. LLMs fail in non-obvious ways, confident hallucinations, context-dependent errors, failure modes that are difficult to anticipate. In low-stakes settings this variance is tolerable. In high-stakes settings the downside tail is too costly to accept even if average performance is good. This suggests that improvements in average LLM accuracy alone may be insufficient to unlock adoption in high-COE occupations, what is needed is a reduction in error variance and the ability to predict reliably when the model will fail. We flag the relative contribution of expected error cost versus error variance as a priority question for future research, and one where interpretability and uncertainty quantification tools may prove decisive.

An Illustration

The intuition is perhaps clearest in radiology, the occupation that first motivated this hypothesis. A radiologist reads medical images to detect tumors, fractures, and bleeds. The consequence of a missed diagnosis is severe: delayed treatment, patient harm, and significant legal liability. AI imaging tools have demonstrated impressive average performance on radiology benchmarks, in some studies matching average radiologist accuracy. And yet clinical adoption remains cautious and heavily supervised, not because radiologists doubt average AI performance, but because they cannot yet predict when the AI will be wrong. A radiologist who countersigns an AI read bears full professional and legal responsibility for it. Until the failure modes are predictable, the rational response is to maintain human oversight of every case, using AI as an augmentation tool rather than a automated one.

What makes this contrast sharper is that the radiologist, as a human expert, solves precisely the problems that make LLM delegation difficult in high-COE settings. Years of medical education, board certification, and supervised clinical training produce a practitioner whose error distribution is relatively predictable, not error-free, but bounded and legible. When a radiologist misses a finding, the failure mode is generally understandable: a subtle presentation, an atypical case, fatigue. The profession has built institutional infrastructure around this predictability, malpractice liability, peer review, credentialing, and continuing education, that creates accountability for errors after the fact and incentivizes error reduction before it. Critically, a radiologist can also explain her reasoning. She can articulate why she read an image as benign, which structures she examined, and what features informed her judgment. This interpretability is not incidental, it is what makes oversight, audit, and accountability tractable.

Data and Measurement

We use Release 4 (Jan 2026) of Anthropic's AI Economic Index dataset following release of Sonnet 4.5. Anthropic publishes task-level usage shares separately for API traffic and Claude.ai traffic. API traffic reflects programmatic usage, firms and developers sending requests via Anthropic's API, typically as part of automated workflows or production applications. Claude.ai traffic reflects direct human usage through Anthropic's consumer interface. Both datasets report each task as a share of total platform traffic, where tasks are defined using O*NET task statement language and classified by Anthropic's internal task classifier.

Approximately 11.7 percent of API traffic is classified as "none" or "not classified", tasks the classifier could not assign to an O*NET statement. Our adoption measure is therefore built on

the classified 88.3 percent of API traffic and represents a conservative lower bound on true penetration.

Task Automation Penetration. Our core adoption measure is constructed as follows. For each 6-digit SOC occupation, we identify all O*NET task statements assigned to that occupation. We then match those task statements to Anthropic API traffic using exact string matching on lowercased task text. Of 2,251 task types observed in Release 4 API traffic, 2,180 (96.8 percent) matched at least one O*NET task statement, confirming that Anthropic's classifier draws directly from O*NET task language.

Task automation penetration for occupation *i* is then:

Adoption_i = (Weighted API task matches_i) / (Total O*NET tasks_i)

The denominator, total O*NET tasks per occupation, is drawn entirely from the O*NET task statement database and is independent of Anthropic data. Using an independent denominator eliminates the selection bias that would arise from measuring API share within API traffic alone, where all tasks in the denominator have already been screened for automation-suitability by the firms sending them.

For tasks that appear in multiple occupations, which describes 97.6 percent of matched tasks, most of which appear in only one occupation, we apply Anthropic's equal-weighting convention: each task contributes 1/N to each of the N occupations it is assigned to. This prevents tasks shared across many occupations from artificially inflating adoption rates in any single occupation.

Occupations with zero API task matches are retained in the analysis as true zeros rather than missing values. Of 770 6-digit SOC codes in our O*NET universe, 275 (35.7 percent) have zero matched API tasks in Release 4. These cluster heavily in production (SOC 51), construction (SOC 47), installation and repair (SOC 49), and transportation (SOC 53), physically embodied occupations where API-based task automation is structurally limited.

Consequence of Error. We use O*NET's Consequence of Error work context variable, which is rated on a 1–100 scale based on surveys of occupation incumbents. The scale captures how serious the consequences of errors are in a given occupation, ranging from low-stakes clerical mistakes to life-threatening professional errors. We average COE scores across O*NET sub-codes within each 6-digit SOC code.

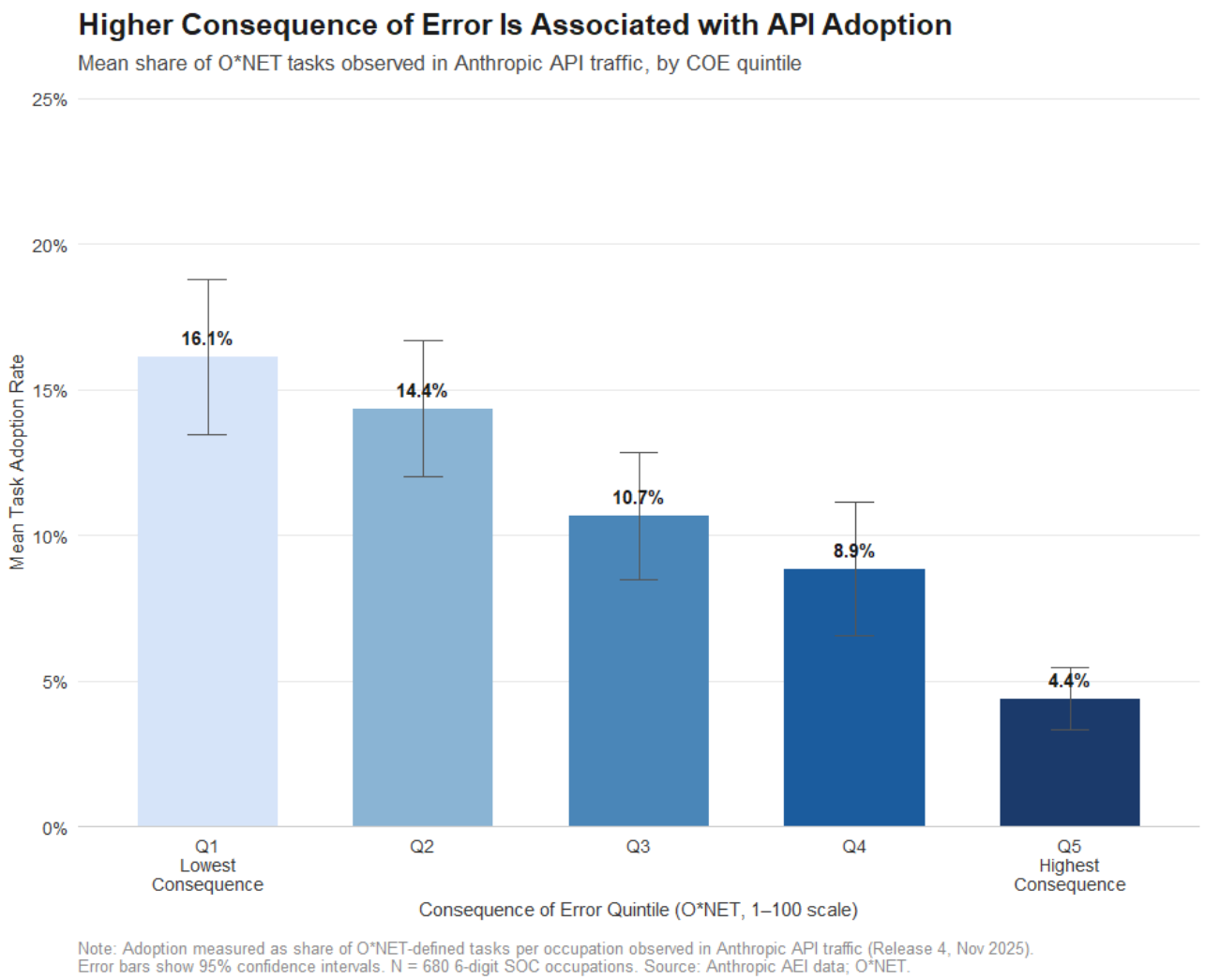
A taxonomy mismatch required an additional processing step. O*NET task statements in our data use 2010 SOC codes, while the O*NET COE file uses 2018 SOC codes. We apply the Bureau of Labor Statistics 2010-to-2018 SOC crosswalk to map COE scores to 2010 codes. For 2010 codes that map to multiple 2018 codes, 42 occupations, including Software Developers, we average the COE scores of the target 2018 codes. Of 770 occupations in our task universe, 680 receive direct COE matches and 80 receive crosswalked COE matches, leaving 10 occupations unmatched. The final analytical dataset contains 760 6-digit SOC occupations.

Without this crosswalk, Software Developers (15-1132), the occupation responsible for approximately 10 percent of all Anthropic API traffic through the single task "modify existing software to correct errors", would have been excluded from the analysis entirely. The

crosswalk assigns Software Developers a COE score of 44.5, reflecting the average of their two 2018 SOC equivalents (15-1252 and 15-1253). This is a moderate score consistent with the nature of software errors, consequential but rarely immediately life-threatening.

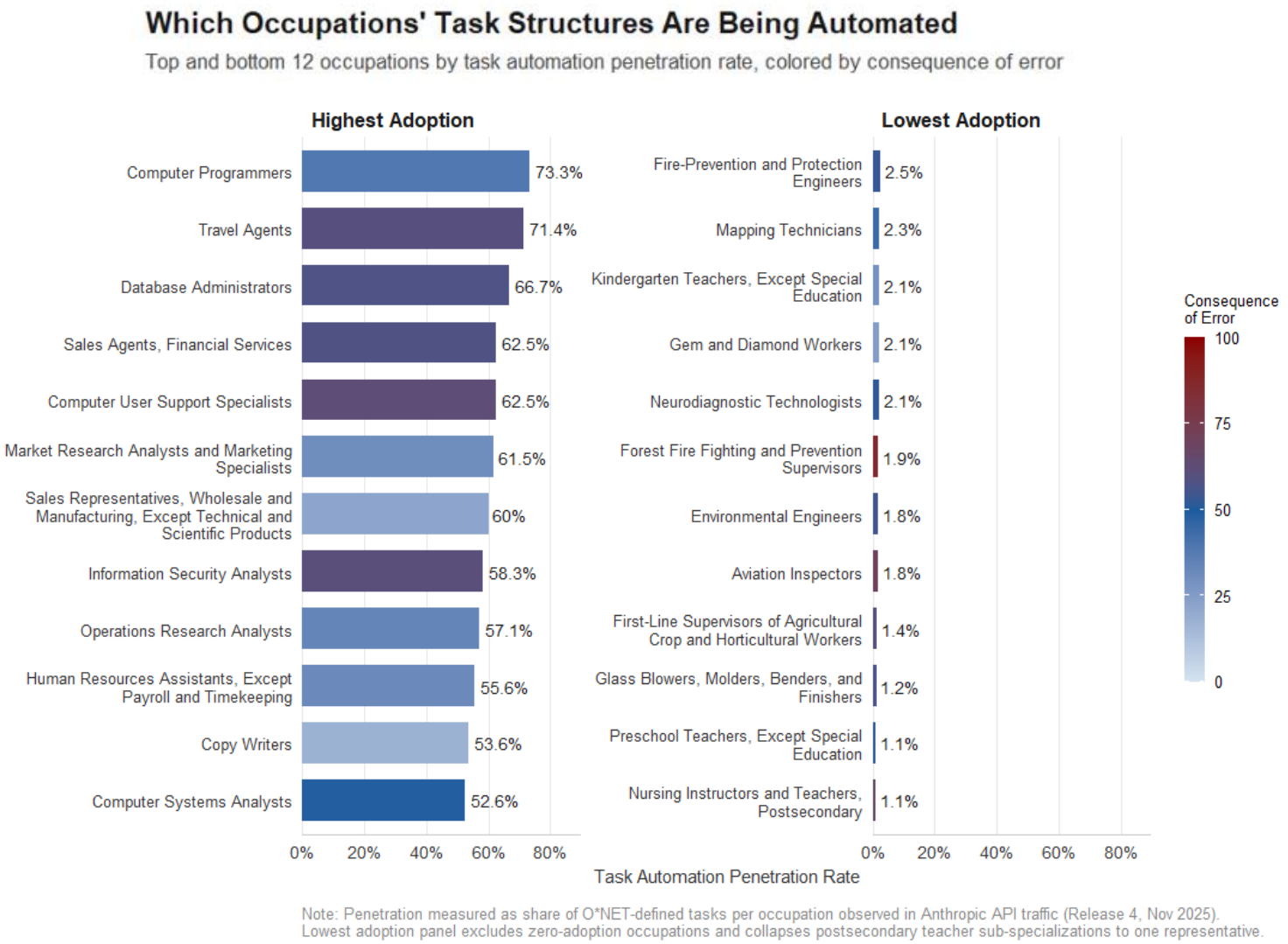
Employment Data. Major group employment figures from the Bureau of Labor Statistics Occupational Employment and Wage Statistics (May 2023) are used for bubble sizing in scatter plots only. They play no role in the analytical weighting.

Results



The COE Gradient Is Monotonic and Steep. Dividing occupations into quintiles by COE score, mean task automation penetration falls monotonically from Q1 to Q5 across the full range of the distribution. Occupations in the lowest consequence quintile (COE 9–33) show mean penetration of 16.1 percent. Each successive quintile shows lower penetration — 14.4 percent, 10.7 percent, 8.9 percent — culminating in 4.4 percent for the highest consequence quintile (COE 68–99). The overall Pearson correlation between task automation penetration (presence in API) and COE score is -0.286 (N=760).

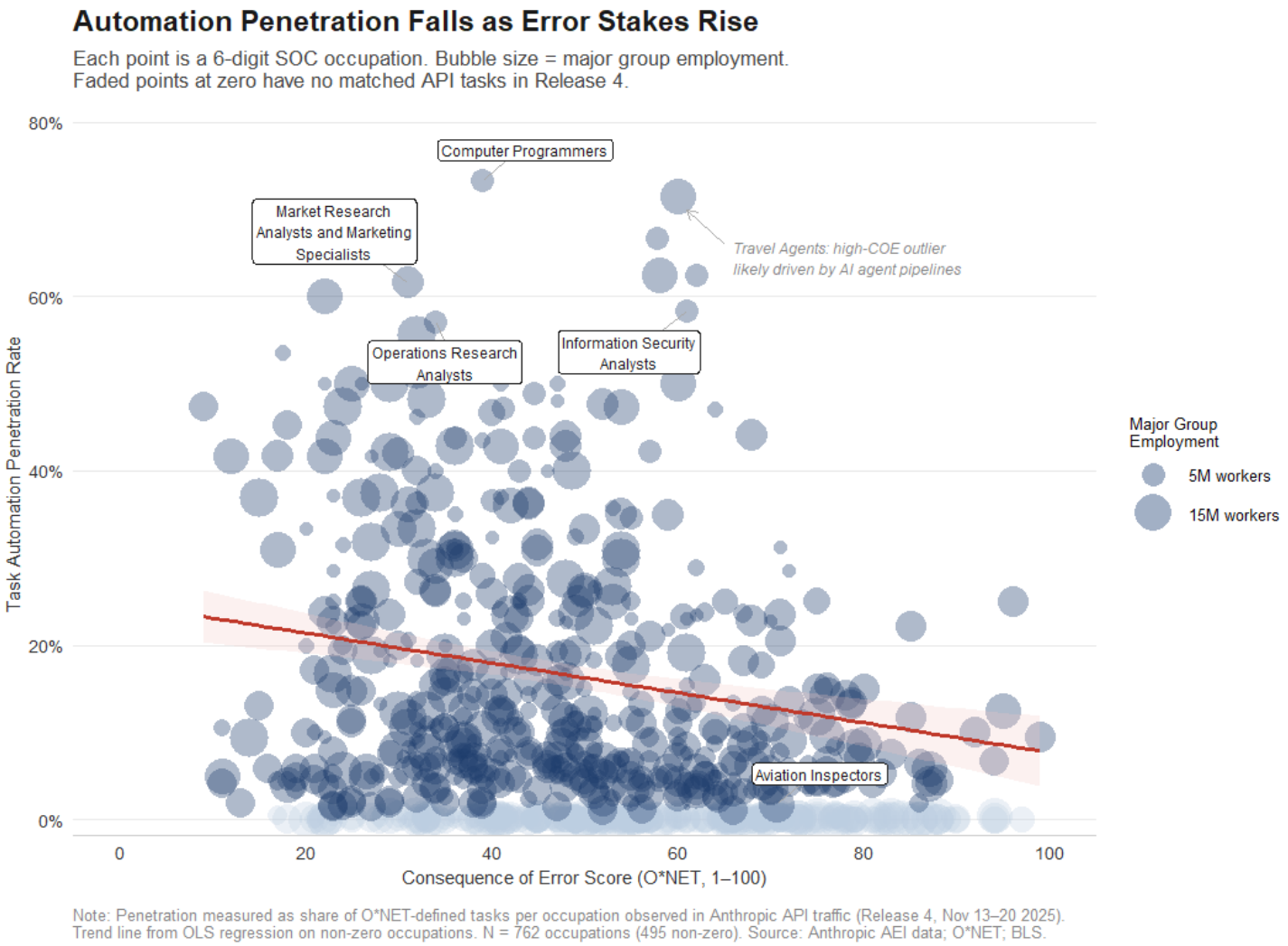
The gradient is visible not just in the quintile means but across the full continuous distribution. The occupation-level scatter shows a clear negative slope with substantial variance, some high-COE occupations show non-trivial adoption rates, and the relationship is not deterministic. But the central tendency is unambiguous.



High-Adoption Occupations Are Concentrated in Low-to-Moderate COE Roles. The highest task automation penetration rates in Release 4 are observed in Computer Programmers (73.3 percent, COE 39), Travel Agents (71.4.5 percent, COE 60), Database Administrators (66.7 percent, COE 57.8), Sales Agents in Financial Services (62.5 percent, COE 58), and Computer User Support Specialists (62.5 percent, COE 62). The task portfolios of these occupations, querying databases, writing and debugging code, researching and synthesizing information, drafting communications, are well-suited to current LLM capabilities and carry moderate consequences when errors occur.

Notably, several high-adoption occupations carry above-average COE scores. Computer User Support Specialists (COE 62), Sales Agents in Financial Services (COE 58), and Information

Security Analysts (COE 61) all appear in the top 10 by adoption rate despite consequence scores above the sample median of 49. This suggests COE is a constraint on adoption at the extremes, particularly at the very high end, but does not perfectly sort every occupation. Task structure, API-friendliness, and the nature of errors in a given occupation all contribute to the adoption decision beyond what COE alone captures.



Low-Adoption Occupations Cluster in Physically Embodied and High-Stakes Roles. Zero or near-zero adoption is concentrated in two distinct occupation types. The first is physically embodied work, production operators, construction trades, transportation workers, agricultural workers, where O*NET task statements describe manual physical actions that cannot be delegated to a language model regardless of error tolerance. The second is high-stakes professional work, surgeons, anesthesiologists, pilots, air traffic controllers, where error consequences are severe and where regulatory, liability, and patient safety considerations create strong barriers to AI delegation even for the information-processing components of the job.

Low adoption occupations fall into two analytically distinct groups. The first comprises high-COE professional and technical roles, Aviation Inspectors, Neurodiagnostic Technologists, Environmental Engineers, where error consequences create a rational brake on delegation

consistent with the paper's central hypothesis. The second comprises physically embodied occupations, Gem and Diamond Workers, Glass Blowers, Agricultural Supervisors, where low adoption reflects structural limits of current LLM technology rather than behavioral responses to error risk. Distinguishing these two mechanisms is important: the first group represents adoption that may increase as model reliability improves and accountability structures develop; the second group represents a more durable barrier that LLMs in their current form are unlikely to overcome regardless of accuracy gains.

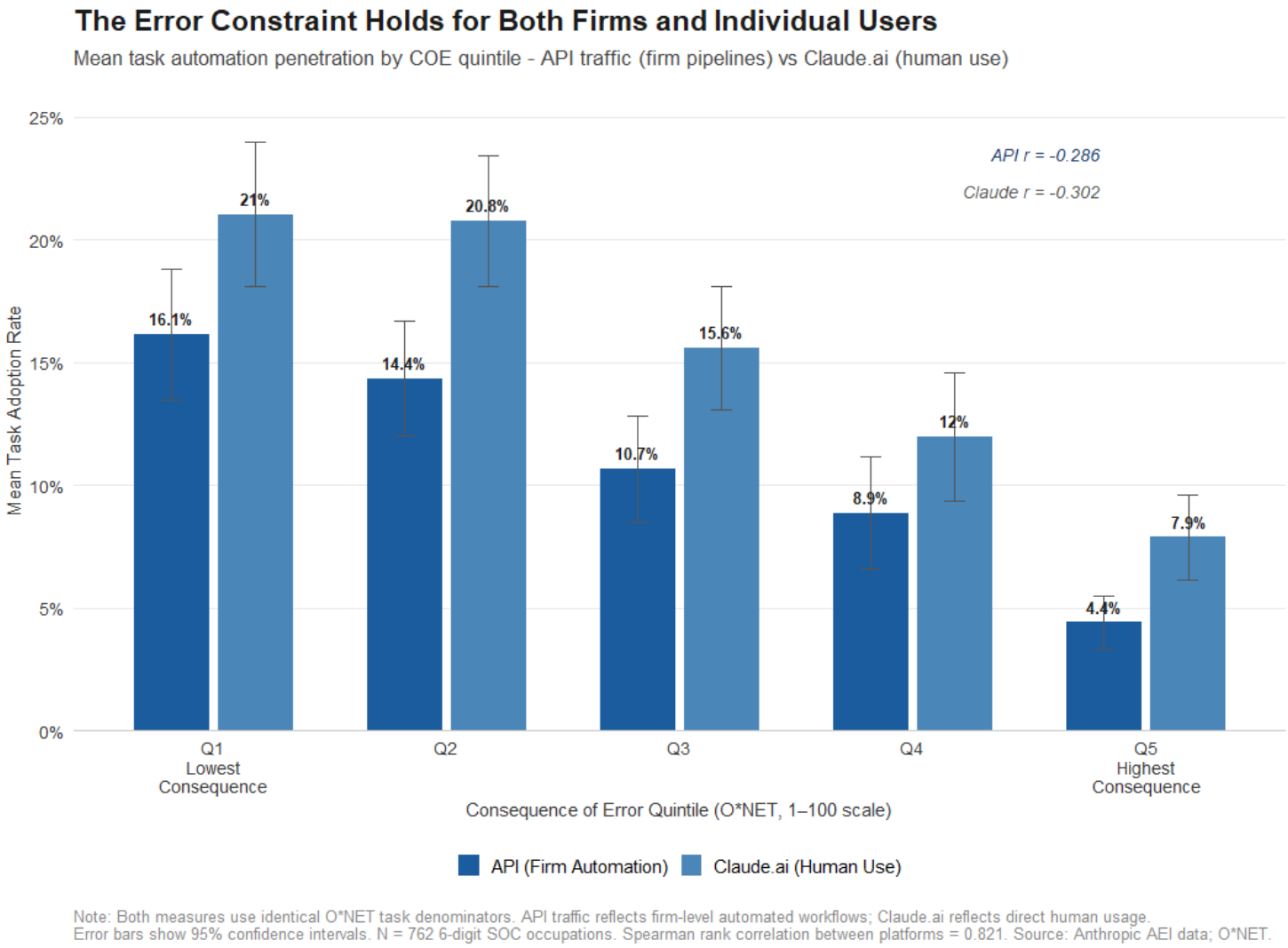
The COE Gradient Is Robust to Platform and Stronger in Human Usage Data. A central interpretive concern with API-based adoption measures is that API traffic in 2025 reflects substantial machine-to-machine communication, AI agents calling Claude as a subcomponent of automated pipelines, rather than human worker behavior. If the COE gradient is entirely a product of developer choices about which pipelines to build, it tells us something about firm automation strategy but less about worker adoption behavior per se.

We address this concern directly using Claude.ai traffic as an independent robustness check. Claude.ai usage reflects direct human interaction with the model and is structurally resistant to AI agent traffic given its pricing model and interface design. Running the identical analysis on Claude.ai task matches produces a COE correlation of -0.302, slightly stronger than the API estimate of -0.286. The quintile gradient in Claude.ai data is proportionally similar to the API gradient, with adoption falling from 21 percent in Q1 to 7.9 percent in Q5.

The Spearman rank correlation between occupation-level API adoption and Claude.ai adoption is 0.825, indicating that the two platforms rank occupations nearly identically. This result has two implications. First, it confirms that the COE gradient is not an artifact of developer pipeline choices, individual human users independently exhibit the same error-constrained adoption pattern. Second, it suggests the two mechanisms reinforce each other: the tasks firms choose to automate and the tasks workers choose to delegate to AI converge on the same low-consequence portion of the task distribution.

Two Mechanisms, One Pattern. Comparing API and Claude.ai adoption by occupation reveals systematic differences that illuminate the underlying mechanisms. Occupations with substantially higher API adoption than Claude.ai adoption, Security Guards, Order Clerks, Telemarketers, Switchboard Operators, Customer Service Representatives, are low-COE, routine, and transactional. Firms are actively building automation pipelines around these tasks, but individual workers do not use Claude.ai for them directly. This is the firm automation channel.

Occupations with substantially higher Claude.ai adoption than API adoption tell the opposite story. Pediatricians show Claude.ai adoption of 54.7 percent against API adoption of just 4.7 percent, a gap of 50 percentage points. Historians, Clergy, Chemical Engineers, and Medical Scientists show similar patterns. These are occupations where individual workers are personally exploring Claude.ai for complex intellectual tasks, but where no firm has yet built automated API pipelines. The high COE scores in many of these occupations (Chemical Engineers: 74, Medical Scientists: 61) are consistent with the hypothesis: firms are deterred from automating these tasks by error consequences, but individual workers making personal decisions about how to use AI in their own practice face a lower threshold.



Limitations

Several limitations qualify the findings and should inform how the results are used.

Cross-Sectional Design. This analysis captures a single week in November 2025. We cannot observe whether adoption is growing, stable, or declining in different occupation groups, or whether the COE gradient is tightening or widening over time. Anthropic has published four releases of AEI data to date, and a panel analysis using all four releases would substantially strengthen causal inference. We treat this as a descriptive cross-section and leave the panel extension to future work.

Task Ownership vs. Worker Adoption. Our adoption measure counts tasks that O*NET assigns to an occupation as appearing in Anthropic traffic, it does not verify that workers in that occupation are the ones sending those API calls. A task assigned to Travel Agents in O*NET may appear in API traffic because a firm is building a travel booking automation product, not because travel agents themselves are using Claude. This distinction matters for interpreting the individual occupation-level results but is less problematic for the aggregate

COE gradient, since task-level automation penetration is itself a labor market-relevant measure regardless of who is doing the automating.

Match Rate and Classifier Dependence. Our analysis rests on Anthropic's internal task classifier assigning API conversations to O*NET task statements. We observe the output of this classifier but not its design. The 96.8 percent match rate gives us confidence that the classifier draws from O*NET task language directly. However, if the classifier systematically over- or under-assigns conversations in certain occupation groups, our adoption measures for those groups will be biased in ways we cannot detect. The 11.7 percent unclassified share of API traffic is a particular concern, if unclassified traffic is concentrated in high-COE occupations, we are understating adoption in exactly the occupations where the hypothesis predicts low adoption, which would artificially strengthen our result.

Equal Weighting Assumption. We follow Anthropic's convention of weighting shared tasks equally across all occupations they are assigned to. A task shared between two occupations contributes 0.5 to each. This is a reasonable default but may misrepresent the true occupational home of a task. A task described as "prepare written reports" is nominally shared across dozens of occupations but is more central to some than others. Task frequency weights from O*NET would allow a more precise attribution.

COE Taxonomy Mismatch. The crosswalk-based COE scores for 80 occupations are averages across 2018 SOC equivalents rather than direct O*NET survey responses. For occupations like Software Developers where the 2010 catch-all code maps to two more specific 2018 codes, the averaged COE score is a reasonable approximation but not a direct measurement. Sensitivity analysis using only the 680 directly matched occupations produces a correlation of -0.296, marginally stronger than the full-sample estimate of -0.288, confirming that the crosswalk does not materially affect the headline result and if anything introduces slight attenuation.

Omitted Variables. COE is correlated with other occupational characteristics that may independently predict adoption. Occupations with high COE tend to require more professional training, operate under stronger regulatory oversight, and involve more interpersonal judgment. Any of these could independently suppress AI adoption. Our cross-sectional design cannot disentangle the COE effect from these correlated channels. We interpret the gradient as consistent with the error-cost hypothesis while acknowledging that it likely reflects a bundle of related constraints of which error consequence is one component.

Implications

The pattern documented here has several implications for how we think about the pace and distribution of AI adoption across the labor market.

First, the COE gradient suggests that the labor market may self-sort in ways that partially protect high-stakes professional work from automation pressure, at least in the near term. Occupations where errors carry serious consequences are precisely those where workers, firms, and regulators have the strongest incentives to maintain human oversight. The data are consistent with this institutional brake operating in practice.

Second, the divergence between API and Claude.ai adoption in high-COE occupations, where human workers explore but firms do not automate, points to a potentially important transition dynamic. Pediatricians using Claude.ai personally to think through cases represents a different kind of adoption than firms building diagnostic automation pipelines. If individual worker experimentation precedes firm-level automation, the high Claude.ai adoption rates in some high-COE occupations may be a leading indicator of future API penetration, or they may reflect a stable equilibrium where personal use and automated deployment remain permanently decoupled by liability and regulatory barriers.

Third, the finding that approximately 35 percent of O*NET occupations show zero API task matches in Release 4 underscores the unevenness of the current automation wave. The occupations with zero penetration are not randomly distributed, they are concentrated in physically embodied work and in the manual trades. For workers in these occupations, the near-term labor market disruption from LLM adoption is likely to be limited not because their work is low-value but because it is structurally resistant to the specific capabilities that current LLMs offer.